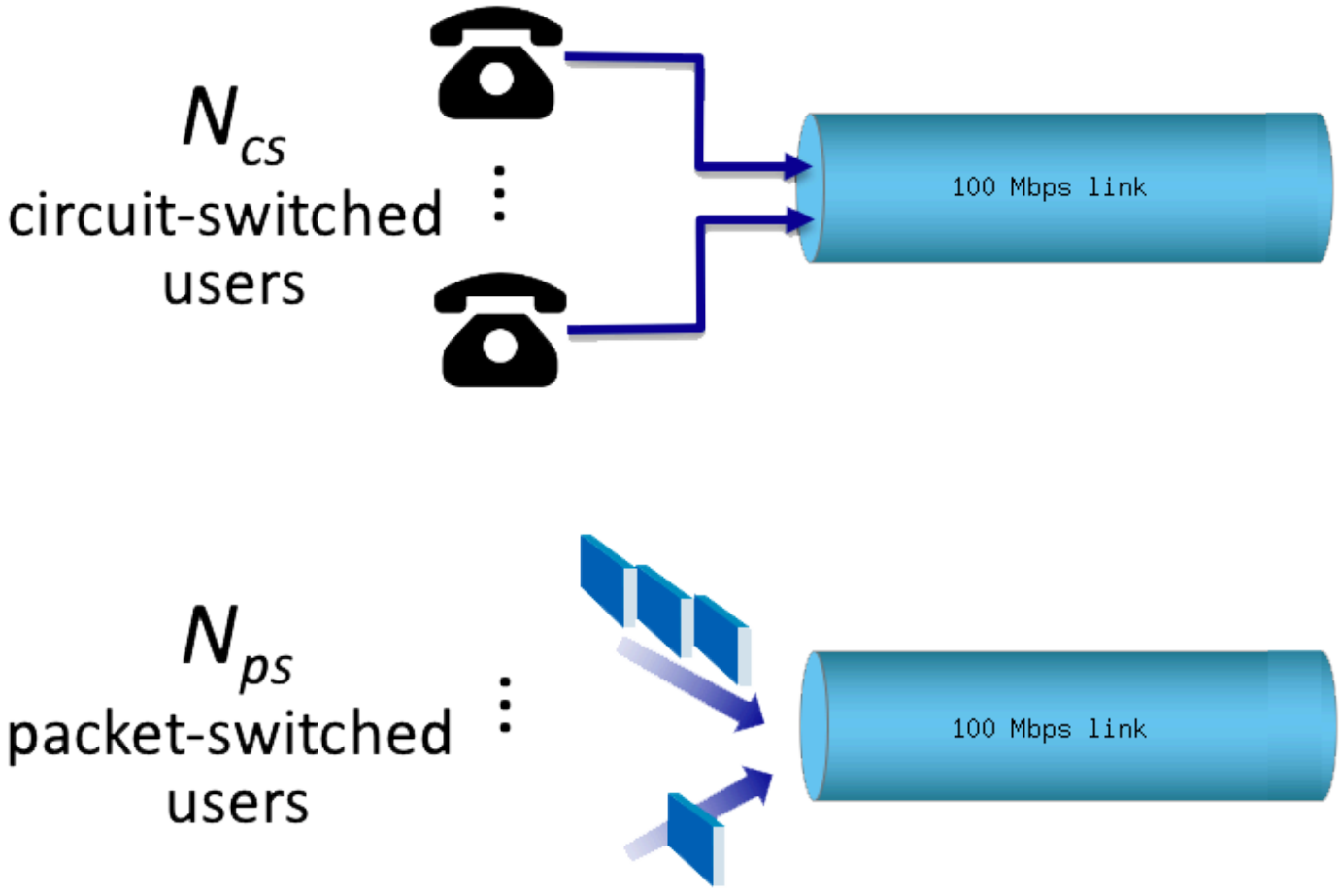
INTERACTIVE END-OF-CHAPTER EXERCISES

QUANTITATIVE COMPARISON OF PACKET SWITCHING AND CIRCUIT SWITCHING

This question requires a little bit of background in probability (but we'll try to help you though it in the solutions). Consider the two scenarios below:

- A circuit-switching scenario in which N_{cs} users, each requiring a bandwidth of 20 Mbps, must share a link of capacity 100 Mbps.
- A packet-switching scenario with N_{ps} users sharing a 100 Mbps link, where each user again requires 20 Mbps when transmitting, but only needs to transmit 10 percent of the time.



Round your answer to two decimals after leading zeros

QUESTION LIST

1. When circuit switching is used, what is the maximum number of users that can be supported?

2. Suppose packet switching is used. If there are 9 packet-switching users, can this many users be supported under circuit-switching? Yes or No.

3. Suppose packet switching is used. What is the probability that a given (specific) user is transmitting, and the remaining users are not transmitting?

4. Suppose packet switching is used. What is the probability that one user (*any* one among the 9 users) is transmitting, and the remaining users are not transmitting?

5. When one user is transmitting, what fraction of the link capacity will be used by this user? Write your answer as a decimal.

6. What is the probability that any 6 users (of the total 9 users) are transmitting and the remaining users are not transmitting?

7. What is the probability that *more* than 5 users are transmitting?

SOLUTION

1. When circuit switching is used, at most 5 users can be supported. This is because each circuit-switched user must be allocated its 20 Mbps bandwidth, and there is 100 Mbps of link capacity that can be allocated.

2. No. Under circuit switching, the 9 users would each need to be allocated 20 Mbps, for an aggregate of 180 Mbps - more than the 100 Mbps of link capacity available.

3. The probability that a given (specific) user is busy transmitting, which we'll denote p, is just the fraction of time it is transmitting, i.e. 0.1. The probability that one specific other user is not busy is (1-p), and so the probability that all of the other N_{ps}-1 users are not transmitting is $(1-p)^{N_{ps}-1}$. Thus the probability that one specific user is transmitting and the remaining users are not transmitting is $p*(1-p)^{N_{ps}-1}$, which has the numerical value of 0.043.

4. The probability that exactly one (any one) of the N_{ps} users is transmitting is N_{ps} times the probability that a given specific user is transmitting and the remaining users are not transmitting. The answer is thus $N_{ps} * p * (1-p)^{N_{ps}-1}$, which has the numerical value of 0.39.

5. This user will be transmitting at a rate of 20 Mbps over the 100 Mbps link, using a fraction 0.2 of the link's capacity when busy.

6. The probability that 6 specific users of the total 9 users are transmitting and the other 3 users are idle is $p^6(1-p)^3$. Thus the probability that any 4 of the 7 users are busy is choose(9, 6) * $p^6(1-p)^3$, where choose(9, 6) is the (9, 6) coefficient of the binomial distribution). The numerical value of this probability is 6.12E-5.

7. The probability that more than 5 users of the total 9 users are transmitting is $\Sigma_{i=6,9}$ choose(9, i) * $p^i(1-p)^{9-i}$. The numerical value of this probability is 6.42E-5. Note that 5 is the maximum number of users that can be supported using circuit switching. With packet switching, nearly twice as many users (9) are supported with a small probability that more than 5 of these packet-switching users are busy at the same time.

QUESTION 1 OF 7

When circuit switching is used, what is the maximum number of users that can be supported?

Answer

CHECK HINT SKIP

We gratefully acknowledge the programming and problem design work of John Broderick (UMass '21), which has really helped to substantially improve this site.

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